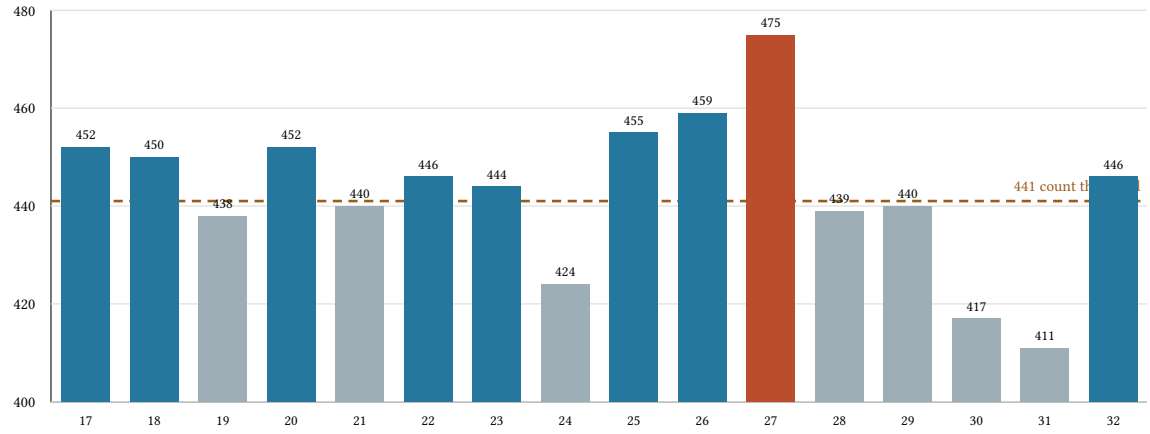
Target-blind 896-element prime coset search

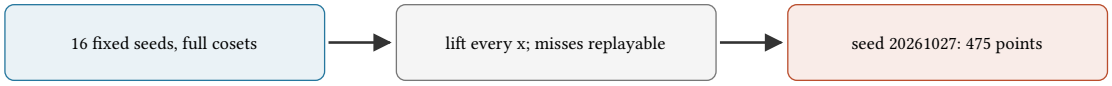
Exact input preparation and independent Sage replay, 2026-10-17 UTC. The 63-bit prime field admits a coset order 896. A frozen 16-seed scan retained every point list and selected only by realized size. Its largest base has 475 points (seed 20261027), above the optimistic m=8 counting threshold of 441 columns. This does not prove target coverage or improve ECDLP cost.

63-bit prime cosets: all 16 frozen seeds

Exact liftable point counts, d=896; target-blind selection by largest realized base



Seed suffix (20261017…20261032); 9 of 16 reach the optimistic count threshold



Dashed threshold is necessary for count capacity; it does not establish coverage, relation yield, or solver speed.

Scan SHA-256: ea1959cee947aa1c740c158250de10c1a73f718b481d5194150f462fa8e395e0

Nine of 16 candidates cross the necessary count threshold. The selected base has signed eight-term multiset upper count 16,944,744,476,780,007,525, or 1.837153 times the subgroup order; collisions can still leave uncovered targets. With seven summands on this same base, uniform-target coverage is at most 1.535760%. Direct S8 therefore remains a seven-summand comparison.

Independent replay and setup

SageMath 10.10.rc0 independently checked 14,336 coset x-values, each stored point, the root orders, and the winner. The complete scan took 18.765831 s wall and 0.829370 s process CPU on this host; constructor-call wall times sum to 0.370852 s and exclude other scan work. Field selection, independent replay, solver, relation, rank, target and rho costs are separate.

Next gate

The prior generic quadratic-lift code locally derives 11 gates per x and a 103-variable, 104-equation, 294-monomial m=8 input for this base. Freeze and independently check that system, then run a bounded direct/lifted comparison against equal-realized-size random and small-x controls on the same targets. Only complete charged relations, rank and unseen target work can establish an end-to-end improvement. The companion Markdown report links all sources, the full scan and the independent replay.